

MSU Graduate Spatial Ecology Lab 4

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Lab 4: Spatial Autocorrelation (Moran's I, Mantel Test)

RMarkdown version of this Lab

This lab has 2 Parts. You need to hand in Part 2 as a **single PDF** produced from R Markdown, including output plots, statistics, code, and text for answers to the questions below. When referring to the code below it may be more useful to use the .Rmd file linked above. OPTIONAL: If you would like to test out GitHub in completing this lab, complete your .Rmd file in RStudio and push your PDF (and your .Rmd if you want) to a repository. You can add the link to the GitHub file when you submit to D2L. Please also submit your PDF to D2L.

This lab uses the brycesite environmental data from the R package labdsv.

The variable list for the site variables is here: <https://www.rdocumentation.org/packages/labdsv/versions/2.1-2/topics/brycesite>. A few notes: east=easting; north=northing; elev is recorded in feet; we will convert to metric.

These data come from a vegetation survey in Bryce Canyon National Park, Utah. It's an amazing place with unique formations! For more about Bryce Canyon see: <https://www.nps.gov/brca/index.htm>

https://upload.wikimedia.org/wikipedia/commons/4/4d/USA_10654_Bryce_Canyon_Luca_Galuzzi_2007.jpg

Part 1: More practice plotting spatial data

Start working in R:

```
##### STARTING UP R
# Clear all existing data (or don't and just make sure you start with a newly opened RStudio session; s
rm(list=ls())

# Close graphics devices
graphics.off()

# Set the paths for your work
output_path<-"output"
# if this folder doesn't exist, create it
if(!dir.exists(output_path)){
  dir.create(output_path)
}

# Create the folders (directories) "data" and "lab4" - If they exist already, this command won't over-w
```

```
data_path<-(file.path("data","lab4"))
if(!dir.exists(data_path)){
  dir.create(data_path,recursive = TRUE)
}
```

If you previously saved your workspace and want to load it here, do so this way:

```
load(file.path(output_path,"lab4.RData"))
```

NOTES on R packages used in this lab:

- (1) With R Markdown, it is helpful to install packages locally before knitting. (copy this into your R console before you knit).

```
for (package in c("sf","dplyr","ape","labdsv","gt","ncf","Feddata","terra","ggplot2","ggspatial","tmap"))
{ if (!require(package, character.only=T, quietly=T)) { install.packages(package) library(package,
character.only=T) } } Load the packages:
```

```
library(sf)
library(dplyr)
library(ape)
library(labdsv)
library(gt)
library(ncf)
library(FedData)
library(terra)
library(ggplot2)
library(ggspatial)
library(tmap)

# Apply ggplot2 theme to remove gray background and set the base font size as 14. See: # https://ggplot.
theme_set(theme_classic(base_size = 14))

# Call the data directly because it's loaded in the labdsv R package
data(brycesite)

# Plot the data in space, by elevation (brycesite has northing and easting so it's projection is UTM).
head(brycesite)
site <- brycesite %>%
  tidyr::drop_na(east, north) %>% # drop missing locations
  mutate(elev_m = elev * 0.3048) # create new column; convert feet to meters

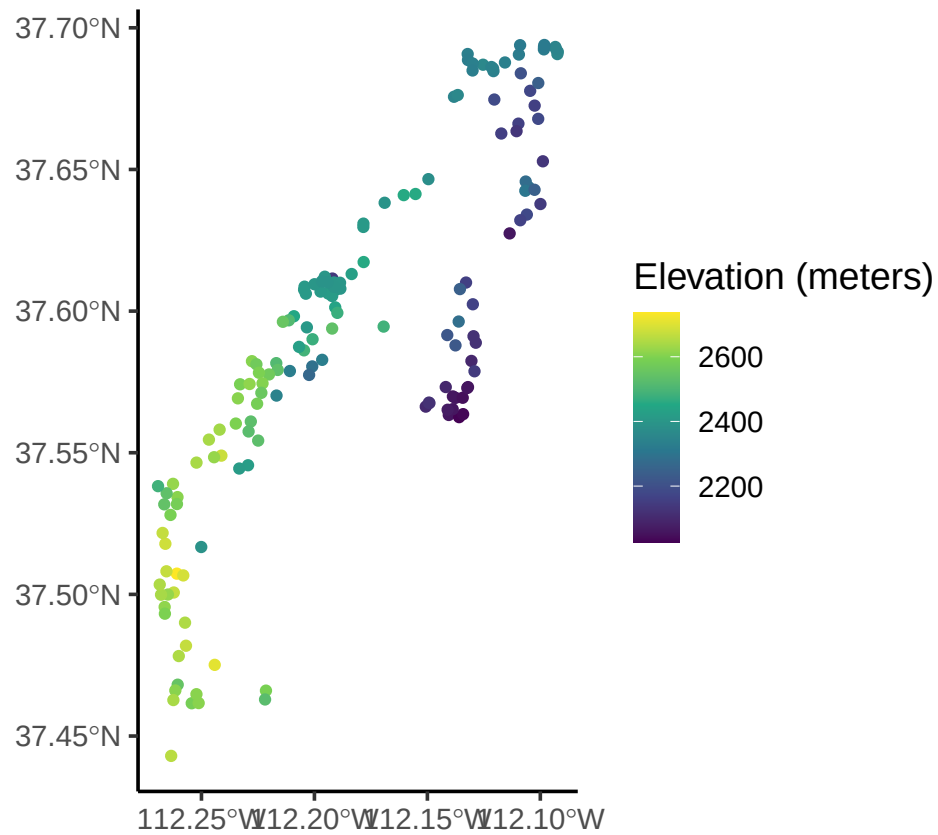
sf::st_crs(site) # CRS for site doesn't exist because it's a dataframe (not spatial feature)

# Define CRS using the sf package
crs.geo <- sf::st_crs("+proj=utm +zone=12 +datum=WGS84")

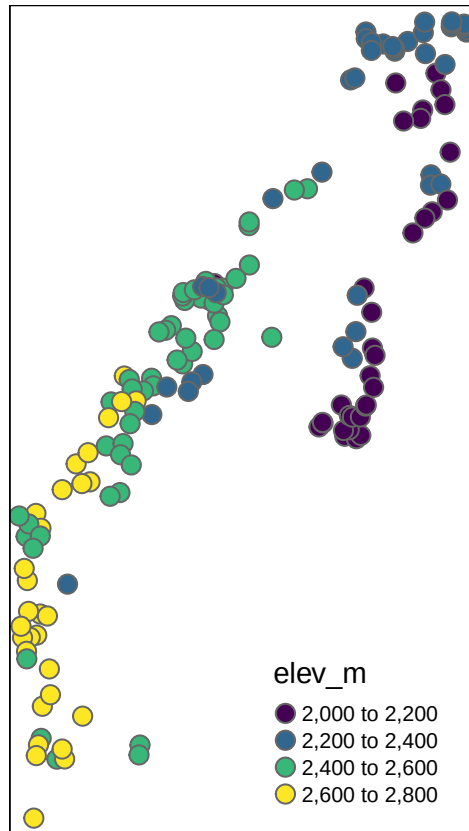
# Make site into a spatial feature using sf package; assign its projection
site <- site %>%
  sf::st_as_sf(coords = c("east", "north"),
  crs = crs.geo)

# Plot it with ggplot
site %>%
```

```
ggplot(aes(color = elev_m)) + geom_sf() +
  scale_color_viridis_c(option = "viridis",
    name = "Elevation (meters)")
```



```
# Plot it with tmap
# tmap
tm_shape(site) + tm_bubbles(palette = "viridis",
  col = "elev_m",
  size = 0.25)
```



*# Take a look at the underlying elevation from the National Elevation Dataset (NED)
using the FedData R package. The FedData package works best if you use a spatial
object created with the sp package.*

```
site.sp<- as(site, "Spatial")
```

```
# Returns a raster
NED <- get_ned(
  template = site.sp,
  label = "bryce"
)
```

If you have trouble with FedData package, the Lab4_NED data are on D2L in Lab 4's # section as "La

Plot the elevation raster and the vegetation data plot locations on the same map. Practice using ggplot2 for this plotting. First, project the shapefile (site, which is vector data) using the CRS of the raster (NED). You should avoid projecting raster data, as projecting it affects the cell values.

```
sf::st_crs(NED)
# Define crs using the sf package
site.m <- sf::st_transform(site, crs=terra::crs(NED))
# Create a dataframe out of the NED raster. This is a necessary step for plotting rasters using geom_raster
NED_df1 = as.data.frame(NED, xy = TRUE)
head(NED_df1)
NED_df1 <- NED_df1 %>%
  tidyr::drop_na()
```

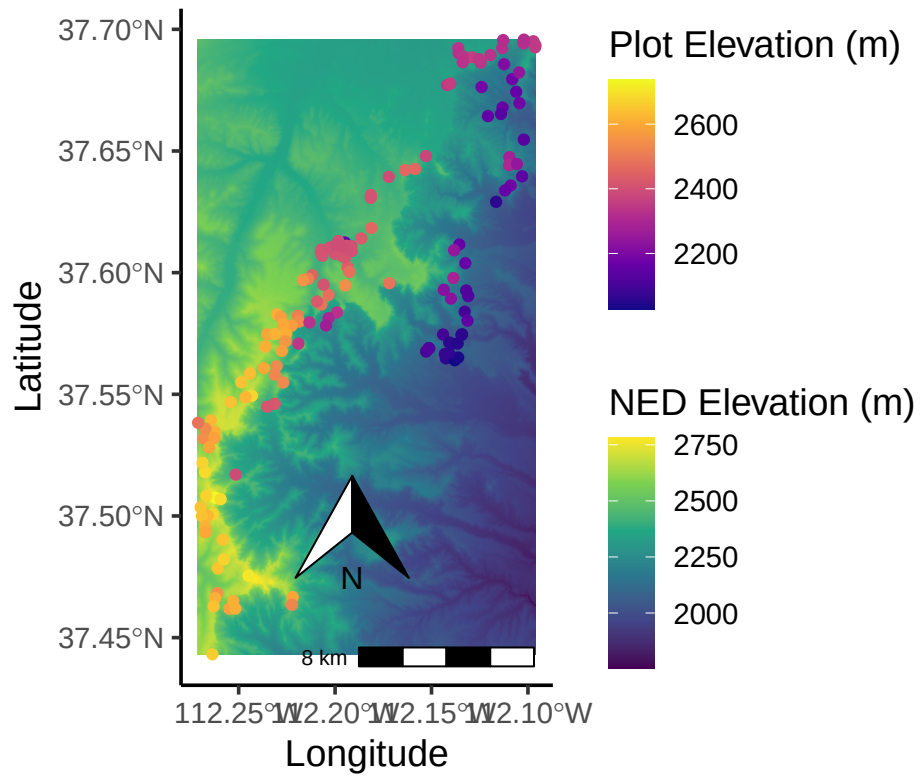
```

# Rename the elevation column
NED_df <- NED_df1 %>%
  rename(elev_NED = USGS_1_n38w113)
# Plot the NED with the points overlaid using ggplot. See how the elevation in the
# points aligns with the NED
NED_elev<-ggplot() +
  geom_raster(data = NED_df,
    aes(x = x, y = y, fill = elev_NED)) # x and y are defined as the column names, "x" # and "y"; "elev_NED"

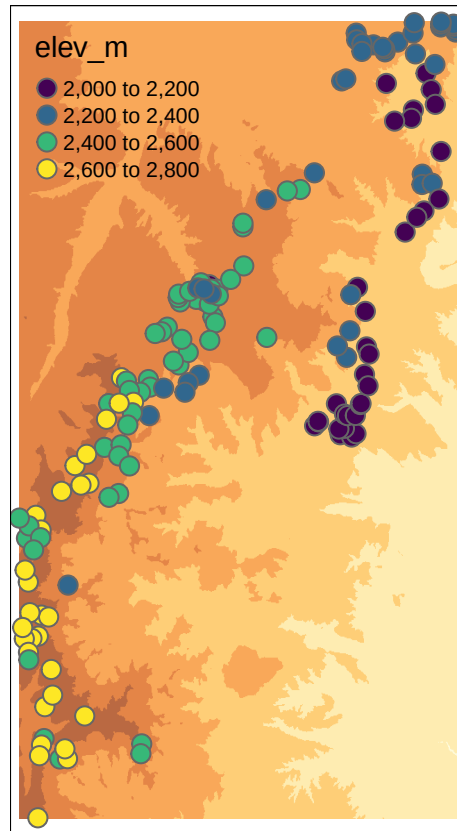
# Now add on the vegetation plots and their associated elevation from the elevdat data,
# and change the color of the raster.
NED_elev <- ggplot() +
  geom_raster(data = NED_df, aes(x = x, y = y, fill = elev_NED)) +
  geom_sf(data = site.m, aes(color = elev_m)) +
  scale_fill_viridis_c(option = "viridis",
    name = "NED Elevation (m)") +
  scale_color_viridis_c(option = "plasma",
    # direction = -1,
    name = "Plot Elevation (m)") +
  annotation_scale(location = "br",
    width_hint = 0.5) +
  annotation_north_arrow(location = "br",
    which_north = "true",
    pad_x = unit(0.75, "in"),
    pad_y = unit(0.5, "in"),
    style = north_arrow_orienteering) +
  labs(title = "Bryce Canyon Vegetation Plot Elevation") +
  ylab("Latitude") +
  xlab("Longitude")
NED_elev

```

Bryce Canyon Vegetation Plot Elevation



```
# Plot with tmap:
tm_shape(NED) +
tm_raster(alpha = 0.75, legend.show = FALSE) +
tm_shape(site.m) + tm_bubbles(palette = "viridis",
                              col = "elev_m",
                              size = 0.25)
```

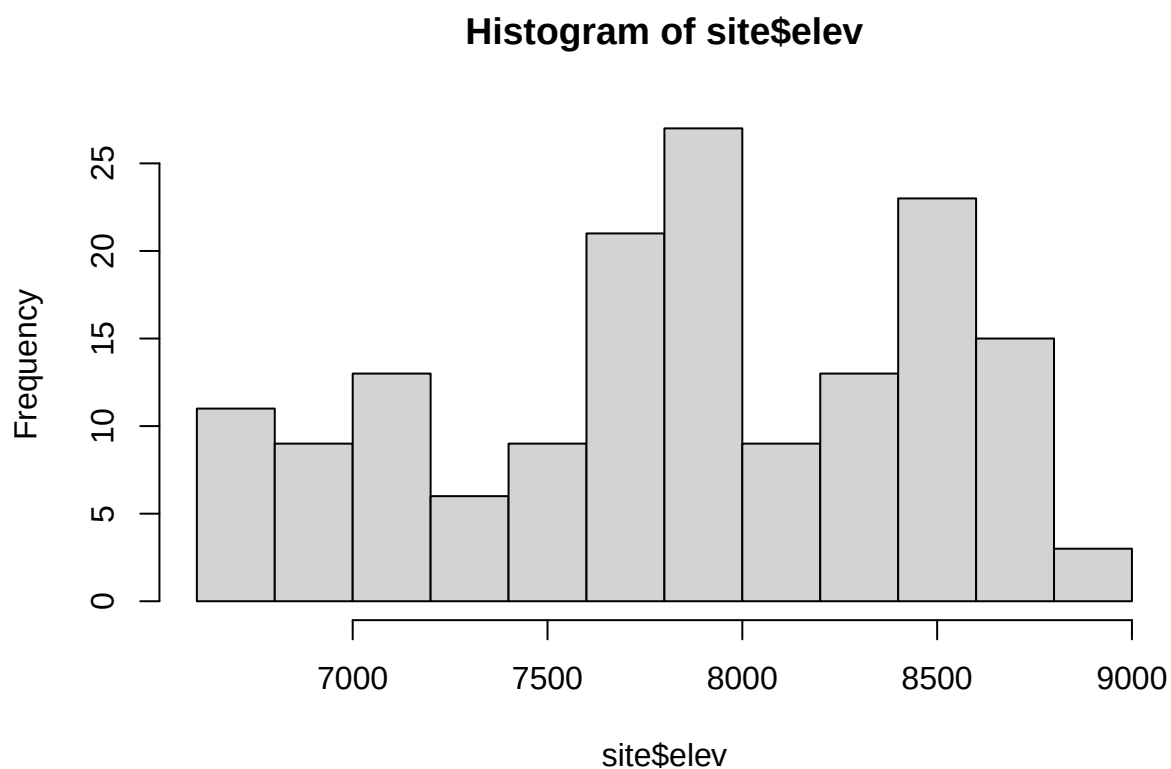


Part 2: Moran's I and Mantel test

Moran's I in package ape

Check assumptions about the elevation data; an assumption of the Moran's I test is that data are normally distributed. Determine whether data should be transformed. Some transformations to consider: square-root, log.

```
hist(site$elev)
```

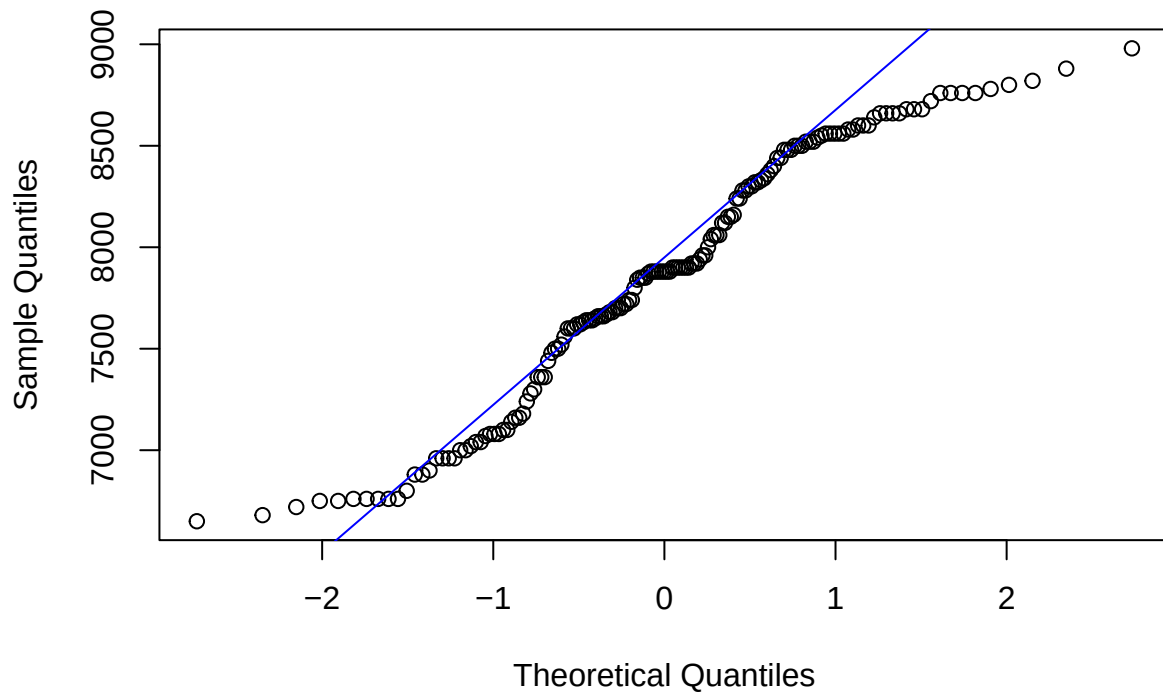


```
shapiro.test(site$elev)
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  site$elev  
## W = 0.95485, p-value = 5.093e-05
```

```
qqnorm(site$elev)  
qqline(site$elev,col="blue")
```

Normal Q-Q Plot



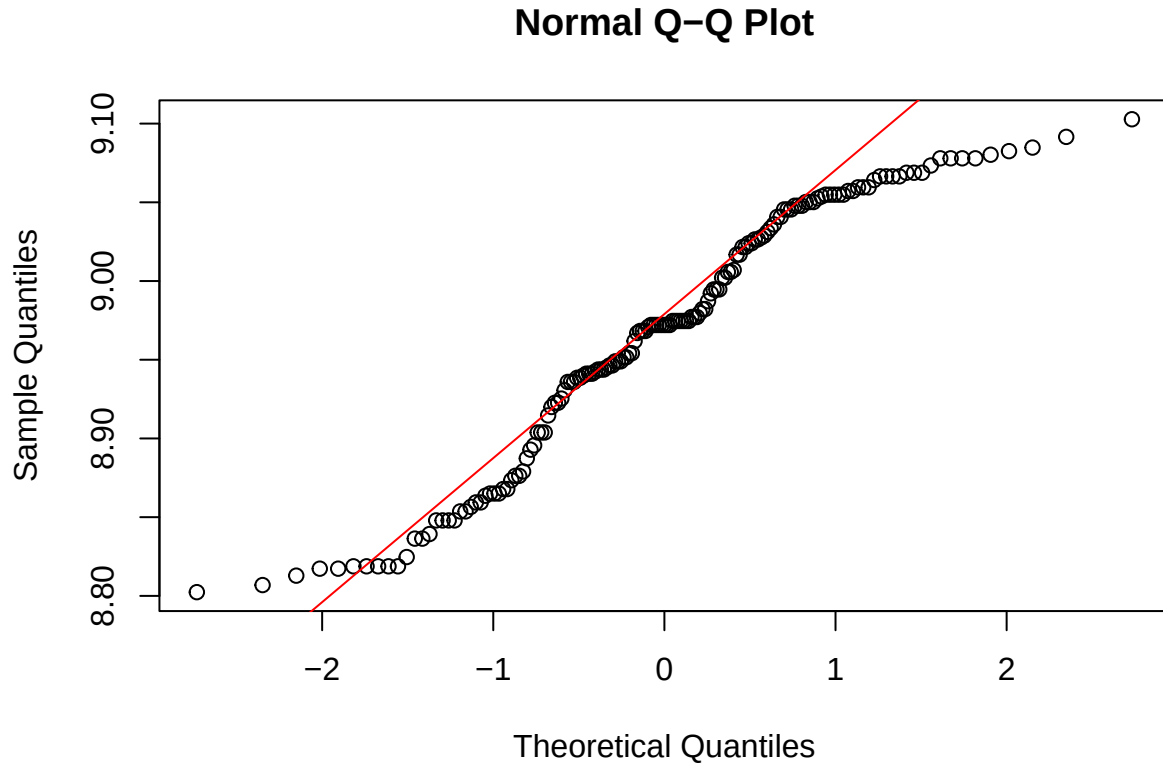
```
shapiro.test(log(site$elev))
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  log(site$elev)  
## W = 0.94937, p-value = 1.689e-05
```

```
qqnorm(log(site$elev))  
qqline(log(site$elev),col="red")
```

Shapiro Test (elevation)

statistic	p.value	method
0.9548468	5.092745e-05	Shapiro-Wilk normality test



```
site$log.elev <- log(site$elev)

# Tidy up the output; use broom and gt packages
shap_test_elev <- broom::tidy(shapiro.test(site$elev_m))
shap_test_elev %>%
  gt() %>%
  gt::tab_header(title = "Shapiro Test (elevation)")
```

```
shap_test_lelev <- broom::tidy(shapiro.test(log(site$elev_m)))
shap_test_lelev %>%
  gt() %>%
  gt::tab_header(title = "Shapiro Test (log elevation)")
```

```
# You can combine them into 1 table like this:
shap_test_df <- data.frame(var = c("elev_m", "log_elev"),
  statistic = c(shap_test_elev$statistic, shap_test_lelev$statistic),
  p.value = c(shap_test_elev$p.value, shap_test_lelev$p.value))
```

Shapiro Test (log elevation)

statistic	p.value	method
0.9493736	1.689473e-05	Shapiro-Wilk normality test

Shapiro Test Results

var	statistic	p.value
elev_m	0.9548468	5.092745e-05
log_elev	0.9493736	1.689473e-05

```
shap_test_df %>%
  slice(1:2) %>%
  gt() %>%
  gt::tab_header(title = "Shapiro Test Results")
```

```
# add easting and northing column to site_sf object
site_coords <- site %>%
  st_as_sf() %>%
  sf::st_coordinates()
site <- site_coords %>%
  bind_cols(site)

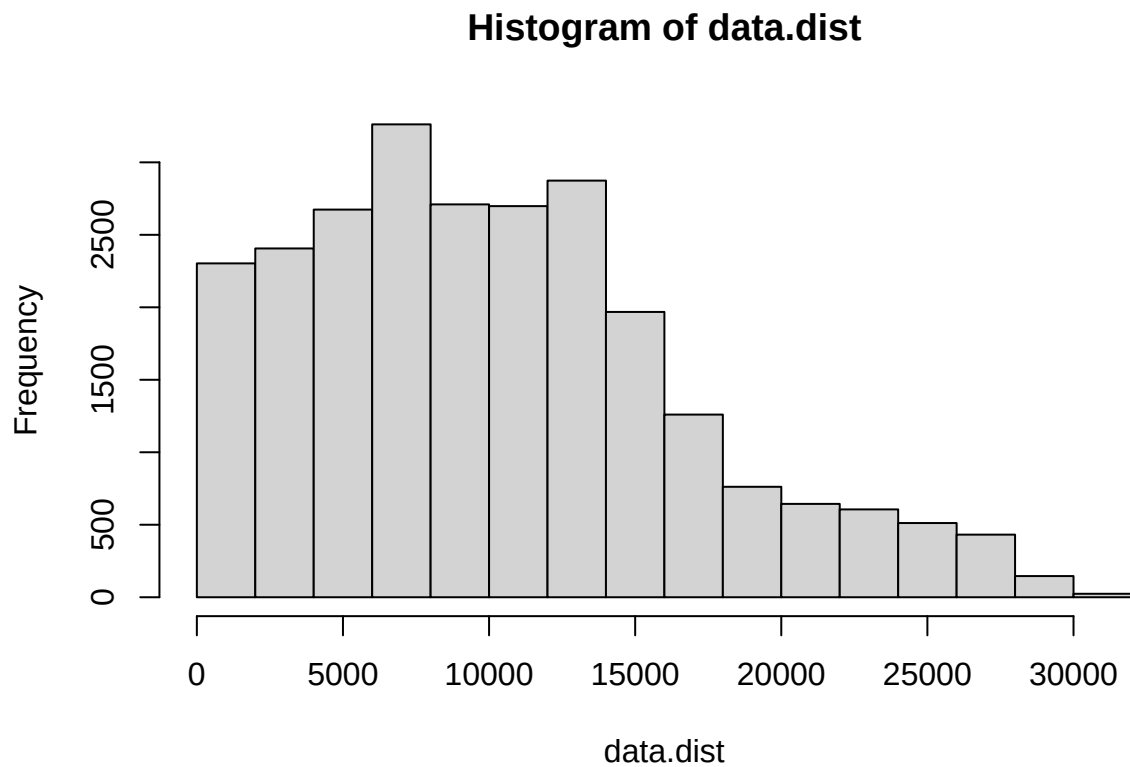
# Compute distance matrix
data.dist <- as.matrix(dist(cbind(site$X,site$Y)))

# Check the maximum distance within the site (units = meters)
max(data.dist)
```

```
## [1] 31583.48
```

```
# Create an inverse distance matrix
w <- 1/data.dist
diag(w) <- 0

# Histogram breaks & counts
hist(data.dist)$breaks
```



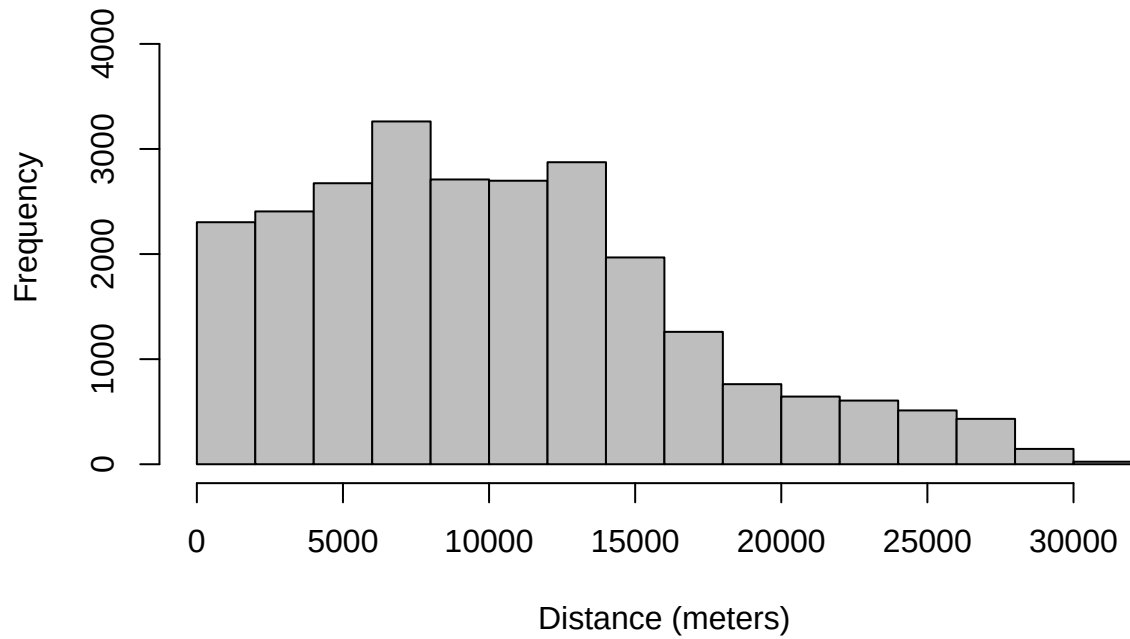
```
## [1] 0 2000 4000 6000 8000 10000 12000 14000 16000 18000 20000 22000
## [13] 24000 26000 28000 30000 32000
```

```
hist(data.dist)$counts
```

```
## [1] 2303 2406 2674 3262 2710 2698 2874 1968 1260 762 644 606 512 432 146
## [16] 24
```

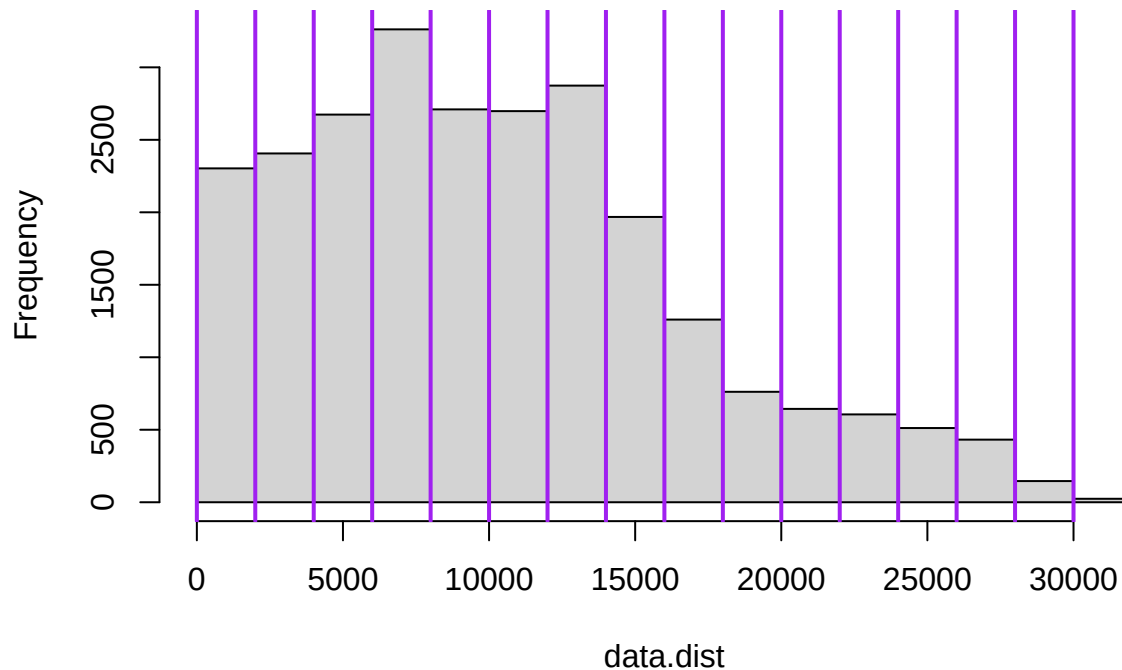
```
# Plot the distance matrix with one option of Distance classes
# (note: these may not be the most informative classes)
hist(data.dist,
      main="Histogram of distances with classes chosen",
      xlab="Distance (meters)",
      col="gray",
      ylim=c(0,4500))
```

Histogram of distances with classes chosen



```
abline(v=hist(data.dist)$breaks,  
       col="purple",  
       lwd=2)
```

Histogram of data.dist



```
#Compute Moran's I
ape::Moran.I(site$log.elev, w, scaled = T, na.rm = T, alternative = "two.sided")
```

```
## $observed
## [1] 0.5036994
##
## $expected
## [1] -0.006329114
##
## $sd
## [1] 0.01934851
##
## $p.value
## [1] 0
```

```
ape::Moran.I(site$elev, w, scaled = T, na.rm = T, alternative = "two.sided")
```

```
## $observed
## [1] 0.5048929
##
## $expected
## [1] -0.006329114
##
## $sd
## [1] 0.0193532
```

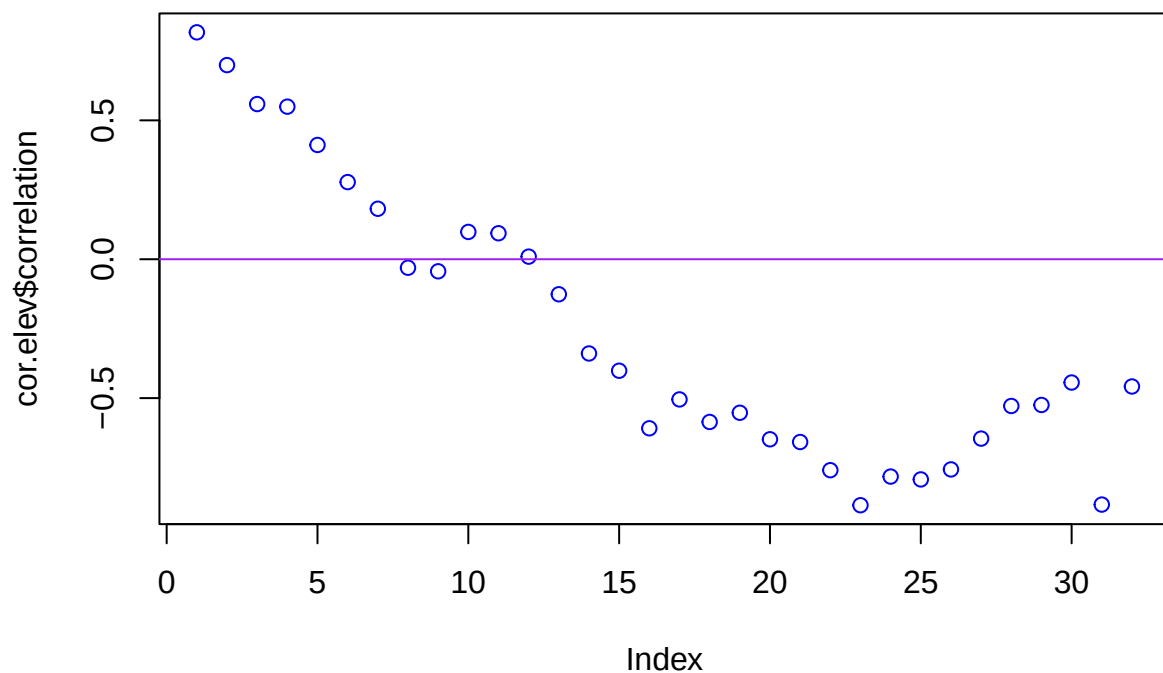
```
##  
## $p.value  
## [1] 0
```

QUESTION 1:

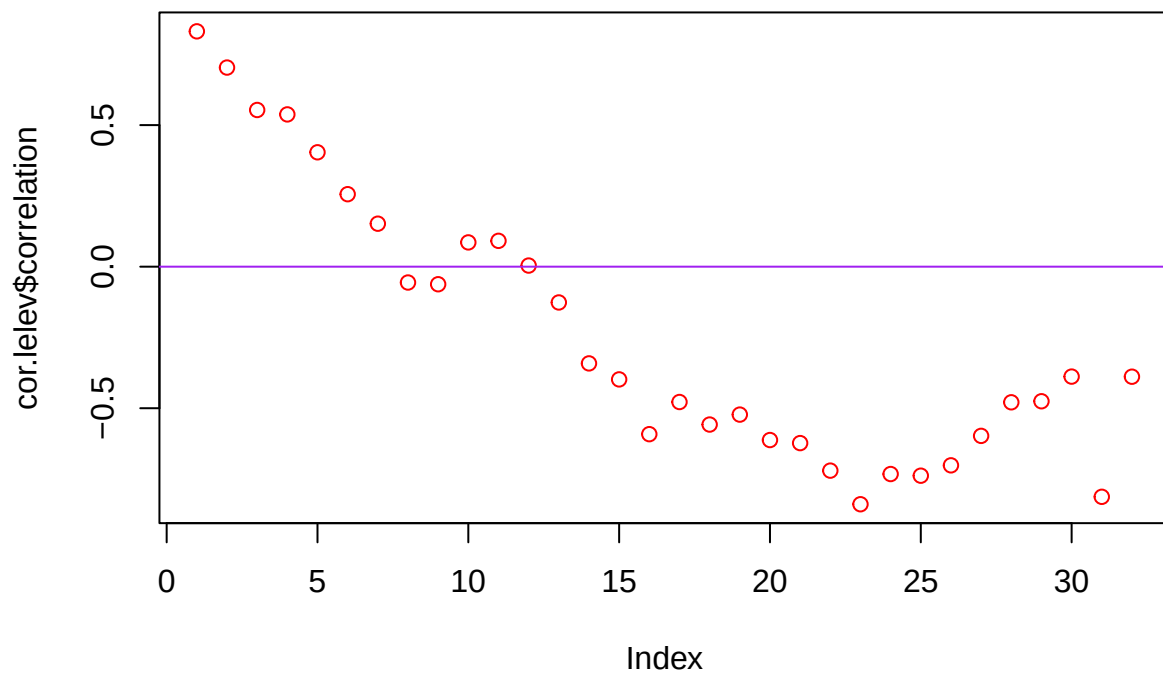
Interpret Moran's I results. Is there a more appropriate form of elevation data (transformed or non-transformed) for this test?

Next, create a Correlogram for Elevation using `correlog` from `ncf` using a specific increment for distance classes (here shown with 1000m increments; use a different distance class in your example).

```
cor.elev <- ncf::correlog(site$X, site$Y, site$elev, increment = 1000, resamp = 100)  
plot(cor.elev$correlation, col = "blue")  
abline(h = 0, col = "purple")
```



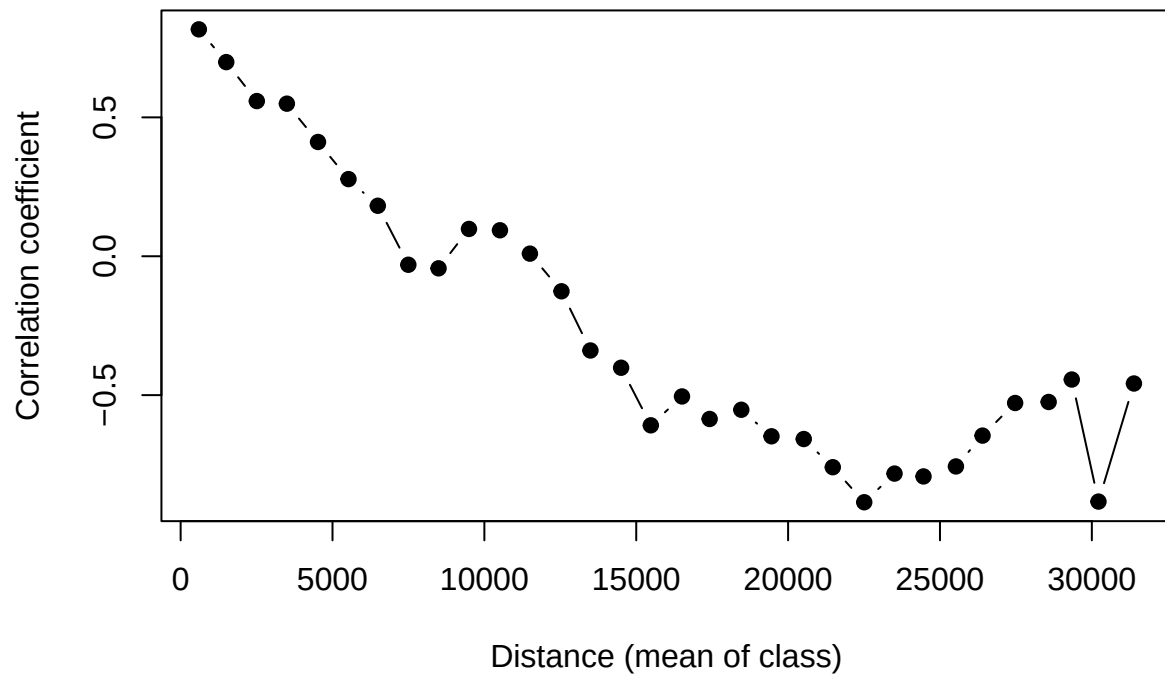
```
cor.lelev <- ncf::correlog(site$X, site$Y, site$log.elev, increment = 1000, resamp = 100)  
plot(cor.lelev$correlation, col = "red")  
abline(h = 0, col = "purple")
```



```
fit <- invisible(cor.elev)
#How many classes are significant by alpha = 0.05?
length(which(fit$p <= 0.05))

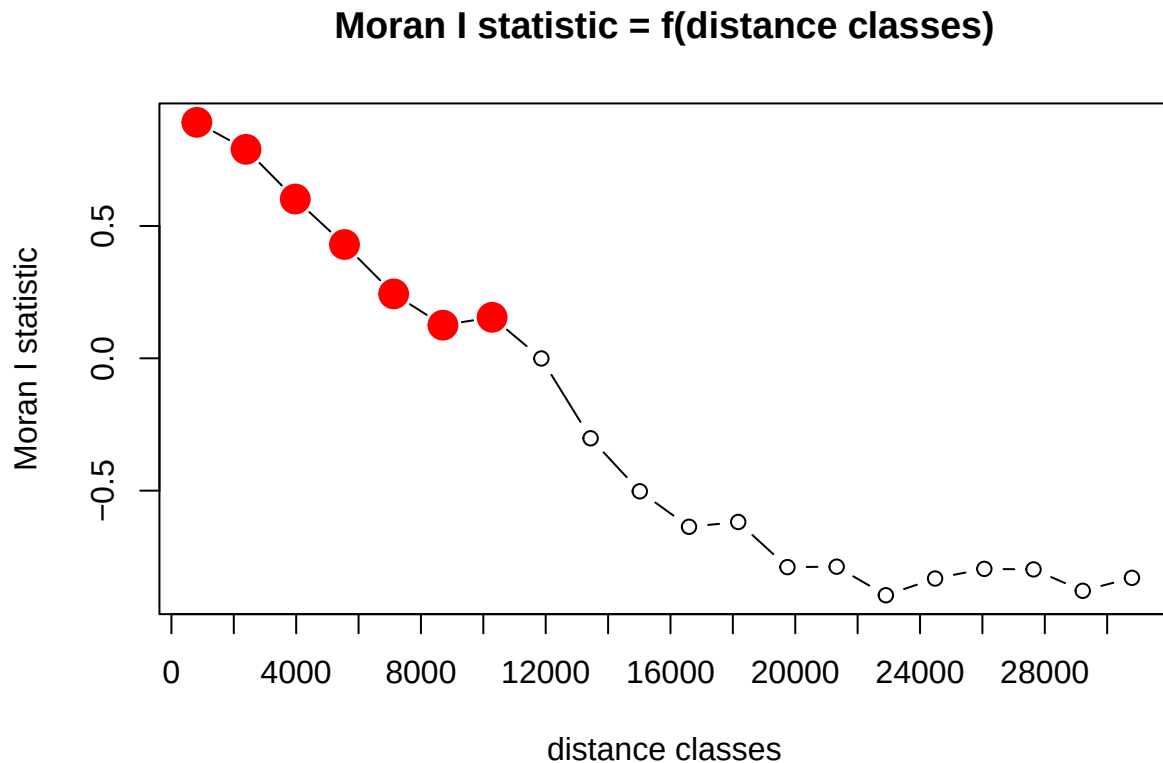
plot(fit$mean.of.class, fit$correlation, pch = 19, "black", ylab = "Correlation coefficient", xlab = "D")
```

Correlogram for elevation change



You can also try other R packages like pgirmess, to make correlograms (note that the # function has to be used as pgirmess::correlog, not pgirmess\$correlog, and the # function has to be used as pgirmess::correlog, not pgirmess\$correlog).

```
pgi.cor <- pgirmess::correlog(coords=site_coords, z=site$elev, method="Moran", nbclass=20)
plot(pgi.cor)
```



QUESTION 2:

Plot and interpret your correlograms. Explain your choice of Distance Classes.

QUESTION 3:

Perform a statistical test to determine if your peaks are significant. Hint: look at your `correlog` output. Which ones are significant?

Mantel Test

Now use the Mantel test to compare differences in elevation at specified points to the distance between those points. This will help you determine whether plots that are closer together have more similar elevations, and if plots that are farther apart have more dissimilar elevations. In this case we're treating the space between sampling locations (their geography) as the predictor variable which is predicting the elevation values.

```
# We already have the euclidean distance matrix (data.dist)
# Compute the distance between elevation measurements
elev.dist <- as.matrix(dist(cbind(site$elev, site$elev)))

# Run a Mantel test on the two matrices
mant.results <- ape::mantel.test(m1 = data.dist, m2 = elev.dist, nperm = 999, alternative = 'two.sided')
```

Next, choose a NEW VARIABLE from the `brycesite` dataframe. Plot the variable in space (see the code for plotting elevation on a map, above). The variable list for the site variables is here: <https://www.rdocumentation.org/packages/labdsr/versions/2.0-1/topics/brycesite>.

QUESTION 4:

What are the general patterns of the NEW VARIABLE across space? Do you expect your NEW VARIABLE to be spatially correlated with elevation across the study region? Why or why not?

QUESTION 5:

Run a Mantel test to look at correlations between the matrices of the two variables (a Mantel test between your NEW VARIABLE distance matrix and `elev.dist`).

You will need to create your `NEWVARIABLE.dist` like this:

```
NEWVARIABLE.dist <- as.matrix(dist(cbind(site$NEWVARIABLE, site$NEWVARIABLE)))
```

Interpret the results of the Mantel tests relative to your predictions from Question 4. You may want to run `correlog()` on your NEW VARIABLE to help with interpretation.

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